

Improving ML Model Accuracy Through Feature Data Configuration and Selection

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Background

Machine Learning & Accuracy

ML models depend on effective feature selection and data configuration for optimal accuracy.

- Feature selection identifies the most relevant data points for prediction.
- Data configuration (train-validation-test splits) impacts generalization and performance

Objectives

- Evaluate how different feature subsets affect prediction accuracy.
- Analyze the impact of different train-validation-test splits.
- Identify the most effective ML model for student performance prediction.

Research Question and Hypothesis

Question: How does feature selection and data configuration impact the accuracy of machine learning models in predicting student academic performance?

Hypothesis: Optimized feature selection and data configuration will improve model accuracy, with **KNN outperforming Linear Regression and Random Forest** for student performance prediction. Due to its ability to capture relationships in categorical and ordinal datasets.

Methods

I. Data Preparation

Student Performance Evaluation dataset from the UCI Machine Learning Repository
Feature Selection: Null Value Preprocessing, Drop Duplication, Type Conversion

II. Testing Machine Models With Test Set Configurations

8:1:1 / 6:2:2 / 7:1.5:1.5

Experiment 1. Linear Regression

Measures linear relationships between features and grades.

Experiment 2. Random Forrest Regression

Uses multiple decision trees for feature importance and non-linear relationships.

Experiment 3. K-Nearest Neighbors (KNN)

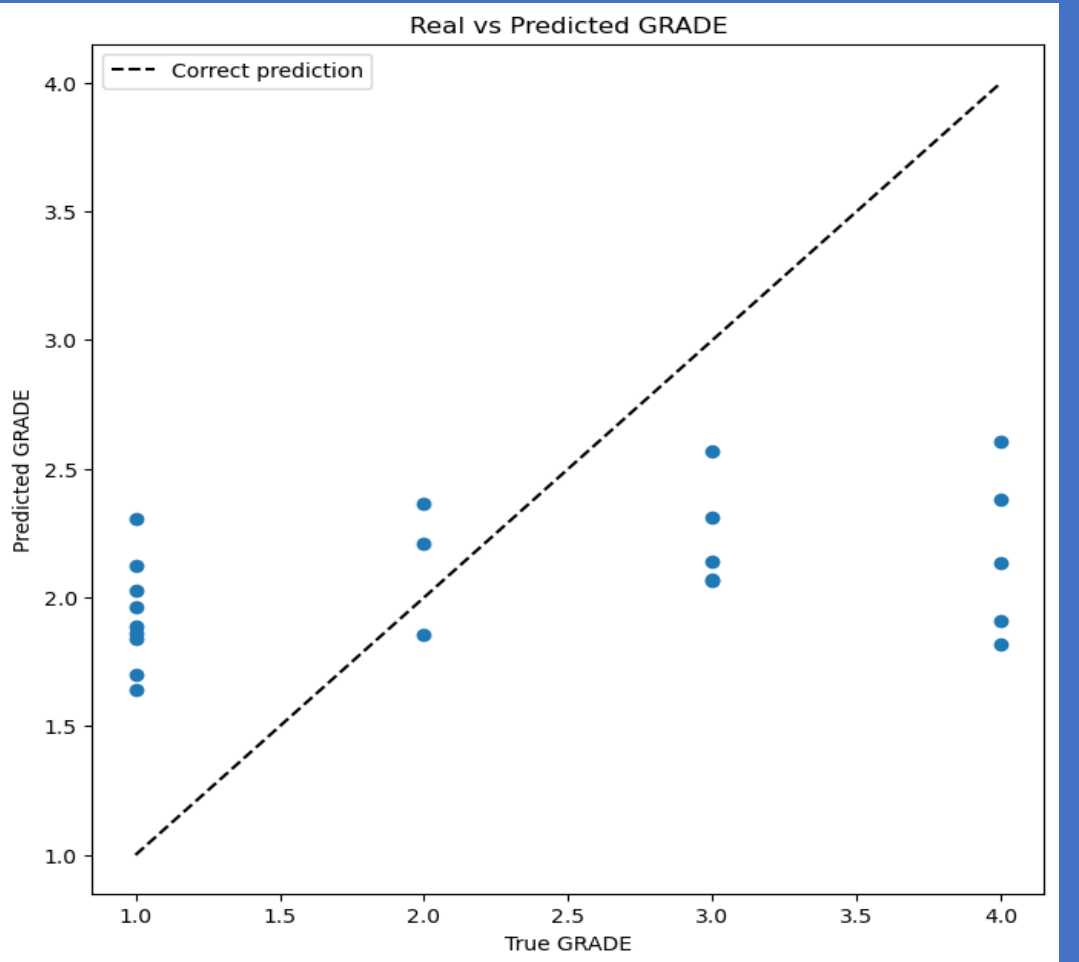
Predicts grades based on similar students' performance using distance metrics.

III. Performance Metrics Analysis (Results)

Results

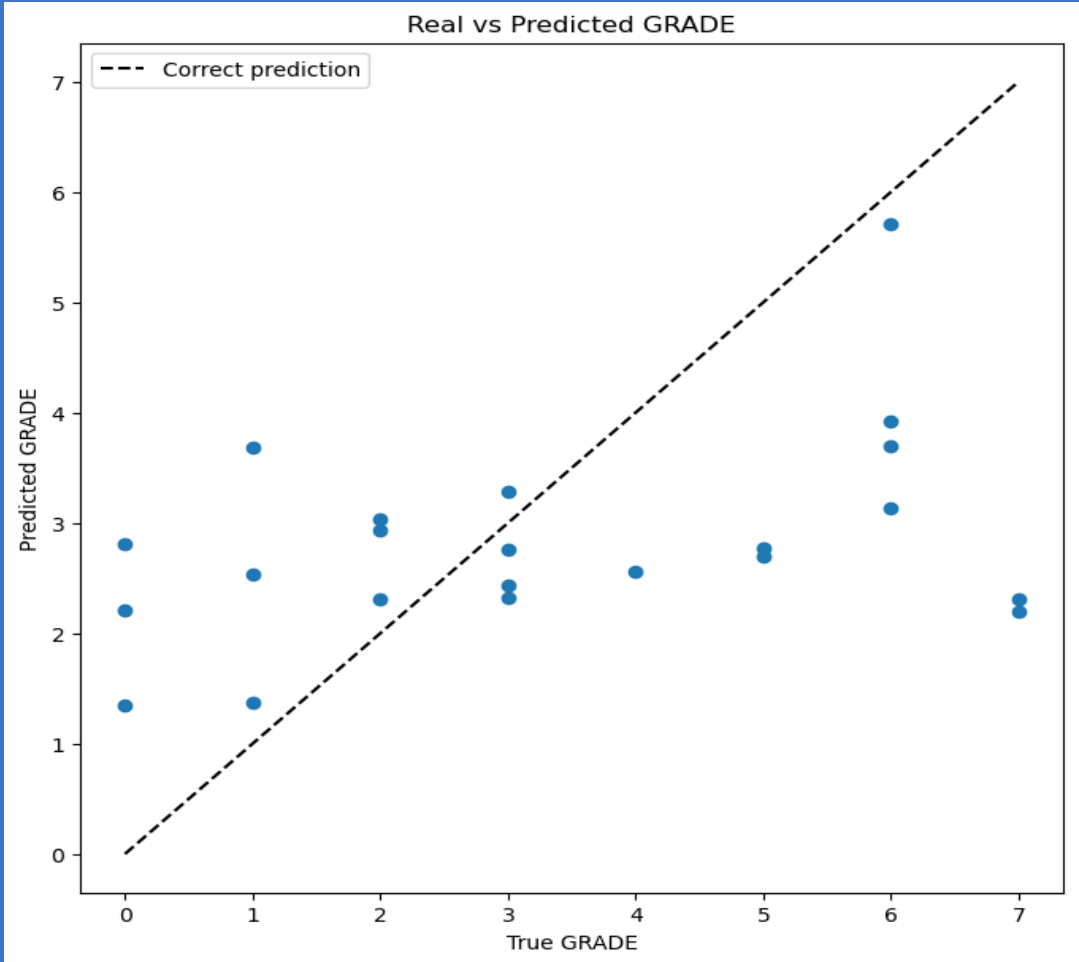
Each model was tested under different feature selection strategies, and their performances were compared to determine the best approach for predicting student academic success.

Performance Metrics For Linear Regression	Train : Validation : Test Data Ratio		
	8:1:1	6:2:2	7:1.5:1.5
MAE(Median Absolute Errors)	1.071	1.151	1.003
MSE(Median Squared Errors)	1.459	1.824	1.300
RMSE(Rooted Median Squared Errors)	1.208	1.351	1.140
R-Squared (Represents the proportion of variance in the dependent variable that is predictable from the independent variable(s). It ranges from 0 to 1.Indicates how well the model explains the variability of the data, 0 ~ 1)	0.062	-0.037	0.116



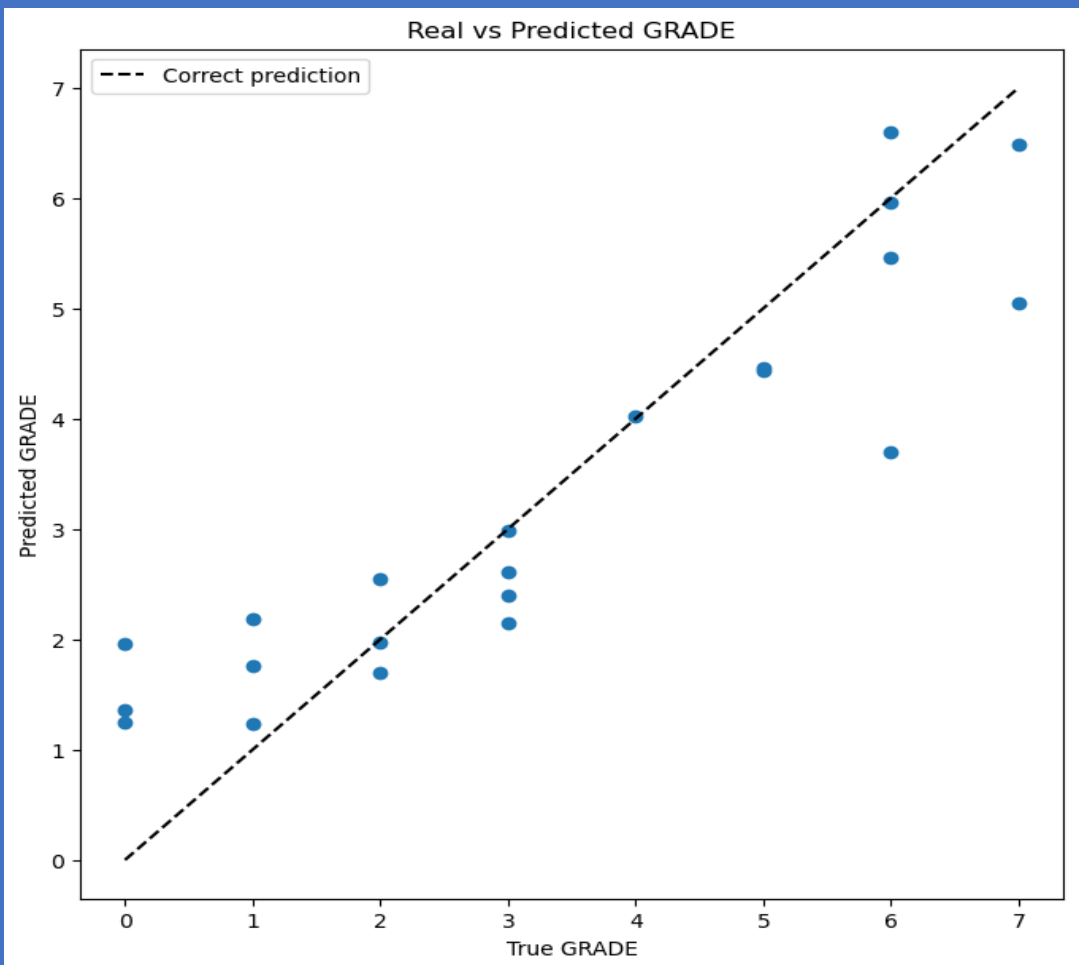
(MAE:1.003) MSE:1.299) (R-squared: 0.116) (RMSE:1.140)

Performance Metrics For RFR	Train : Validation : Test Data Ratio		
	8:1:1	6:2:2	7:1.5:1.5
MAE(Median Absolute Errors)	1.786	1.670	1.727
MSE(Median Squared Errors)	4.249	4.451	4.663
RMSE(Rooted Median Squared Errors)	2.061	2.109	2.159
R-Squared	0.135	0.031	0.121



(MAE:1.727) (MSE:4.663) (R-squared:0.121) (RMSE:2.159)

Performance Metrics For KNN	Train : Valid : Test Data Ratio		
	8:1:1	6:2:2	7:1.5:1.5
MAE(Median Absolute Errors)	0.632	0.723	0.752
MSE(Median Squared Errors)	0.547	0.849	0.978
RMSE(Rooted Median Squared Errors)	0.740	0.921	0.989
R-Squared	K=2, 0.847	K=2, 0.712	K=2, 0.726
	K=3, 0.823	K=3, 0.746	K=3, 0.774
	K=4, 0.889	K=4, 0.777	K=4, 0.815
	K=5, 0.877	K=5, 0.815	K=5, 0.816



MSE: 0.9781 MAE: 0.7522
KNN Metrics: Final Model (K=5, weights=distance) : R²Score: 0.816 RMSE: 0.989

Limitations

We observed that the choice of train-validation-test split ratio has little effect on model performance when working with datasets containing fewer than 1,000 instances. The dataset contains only 145 student records, which may not be sufficient for robust generalization. While the dataset includes various academic and personal factors, it does not account for external influences such as socio-economic background, mental health, or study environments.

Conclusion

We plan to expand our research by exploring other machine learning libraries such as **TensorFlow Keras** and **LightGBM**.

These libraries offer advanced tools and techniques, including deep learning and gradient boosting, which can provide additional perspectives on model performance and capabilities

We are also interested in testing a larger dataset to evaluate whether the predictive performance of the machine learning models changes with an increased data size.

Acknowledgements

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